ASSIGNMENT-4

Full Marks: 70

Q1. [CO1] 5+5

a) **Show** the IV characteristics graph of a MOSFET. Identify the different operating regions in the graphs.

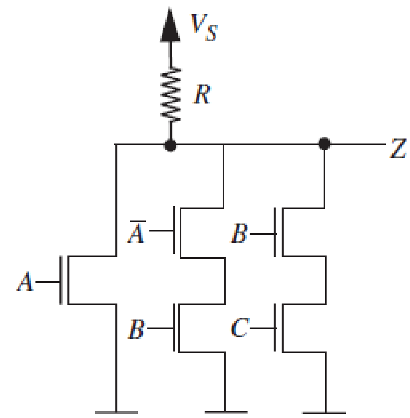
b) **Draw** the VTC of a NOR gate using MOSFET S model.

Q2. [CO2 + CO3] 10+5

a) **Analyze** the circuit-1 to find Z in terms of Boolean inputs A, B, C and write a Truth Table.

b) **Design** a circuit using MOSFETs (S model) to implement the operation specified in following truth table.

A	B	X (output)
0	0	1
0	1	0
1	0	0
1	1	1



Circuit-1

Q3. [CO3] 15

Design a circuit using MOSFETs to implement the following expressions.

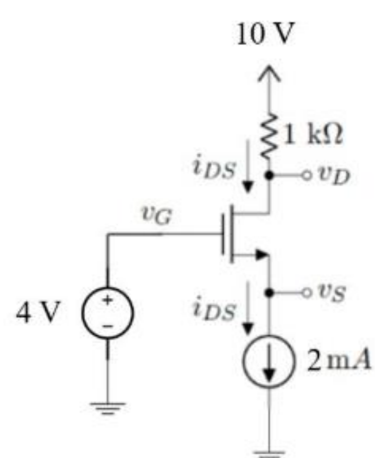
a) f = (A+B)C

b) f = ($\overline{A}$B+CD)

c) f = AB+A+CD

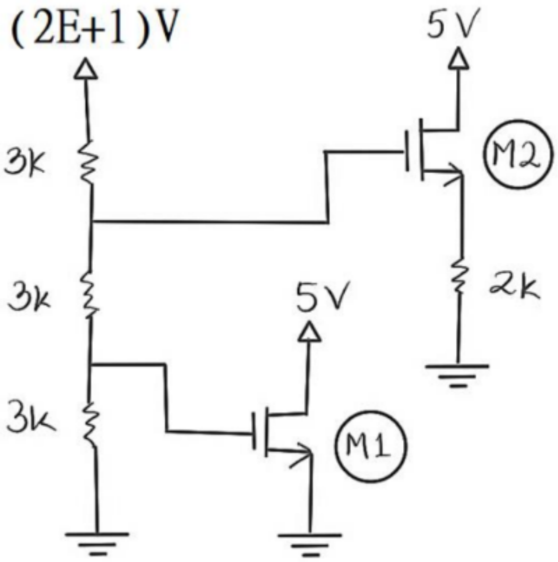
Q4. [CO2] 10

Analyze the circuit to find V_D , use method of assume state. You must validate your assumptions. For the MOSFTET, V_T = 1v and k = 4mA/v^2.



Q4. [CO2] **10**

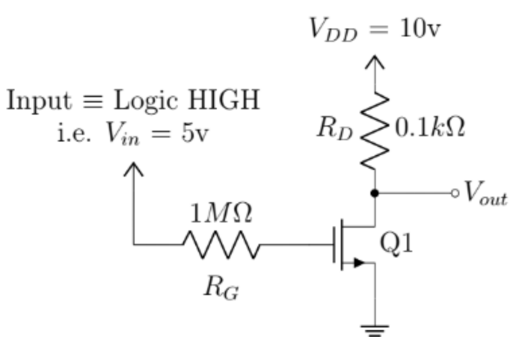
You are given (i) E=2, or, (ii) E=5. Determine the operation modes of the *M1* and *M2* MOSFETs. Also, find the power dissipated in the 2kΩ resistor.



For both mosfets,
$V_T = 0.3\ V$
$k_n = 0.1\ mA/V^2$

Q5. **10**

Joy and Nirmol were designing an inverter circuit with MOSFET for their project. Joy designed the following circuit. However, Nirmol said that the designed inverter circuit would malfunction. When they tested the circuit, it malfunctioned. For logic HIGH input, the output was also logic HIGH instead of logic LOW. Nirmol modified the value of R_D to a larger value, and eventually it fixed the issue.



For Input	
Voltage Level	*Logic Level*
$V_{in} = 0$v	LOW i.e. '0'
$V_{in} = 5$v	HIGH i.e. '1'
For Output	
Voltage Level	*Logic Level*
$V_{out} > 5$v	HIGH i.e. '1'
$V_{out} < 5$v	LOW i.e. '0'

MOSFET Equations

Cut-off Mode

$I_{DS} = 0$

Triode Mode

$I_{DS} = k\left[V_{OV}V_{DS} - \frac{1}{2}V_{DS}^2\right]$

Saturation Mode

$I_{DS} = \frac{1}{2}kV_{OV}^2$

MOSFET Parameters

$V_T = 1v,\ K = 4mA/V^2$

Inverter Circuit of Joy and It's Corresponding Voltage Level, Logic Level

(a) [CO1] **State** the conditions to verify the assumptions regarding the Saturation Mode of a MOSFET. [1]

(b) [CO2] **Analyze** the circuit shown above and **calculate** the value of V_{out} using the method of assumed states. You must **validate** your assumption. **Determine** whether Nirmol was correct or not. [3]

(c) [CO2] Nirmol changed the value of R_D to $1k\Omega$. Now, **Analyze** the modified circuit and **calculate** V_{out}. **Determine** whether the the inverter was working properly after the modification. [3]

(d) [CO3] **Design** the inverter circuit by choosing a suitable value of R_D such that, the MOSFET operates at the edge of saturation. [Hint: At the edge of saturation, $V_{DS} = V_{OV}$] [3]